

HW1

Exercise 1. Let $\mathbf{x} = (1, 2, 3)^T$, $\mathbf{y} = (y_1, y_2, y_3)^T$, and $\mathbf{z} = (4, 2, 1)^T$. Compute $2\mathbf{x}$, $3\mathbf{y}$, and $\mathbf{x} + 2\mathbf{y} - 3\mathbf{z}$.

Exercise 2. True or false:

- (a) Every vector space contains a zero vector.
- (b) If f and g are polynomials of degree n , then $f + g$ is a polynomial of degree n .
- (c) If f and g are polynomials of degree n , then $f + g$ is a polynomial of degree less than or equal to n .

Exercise 3. True or false:

- (a)